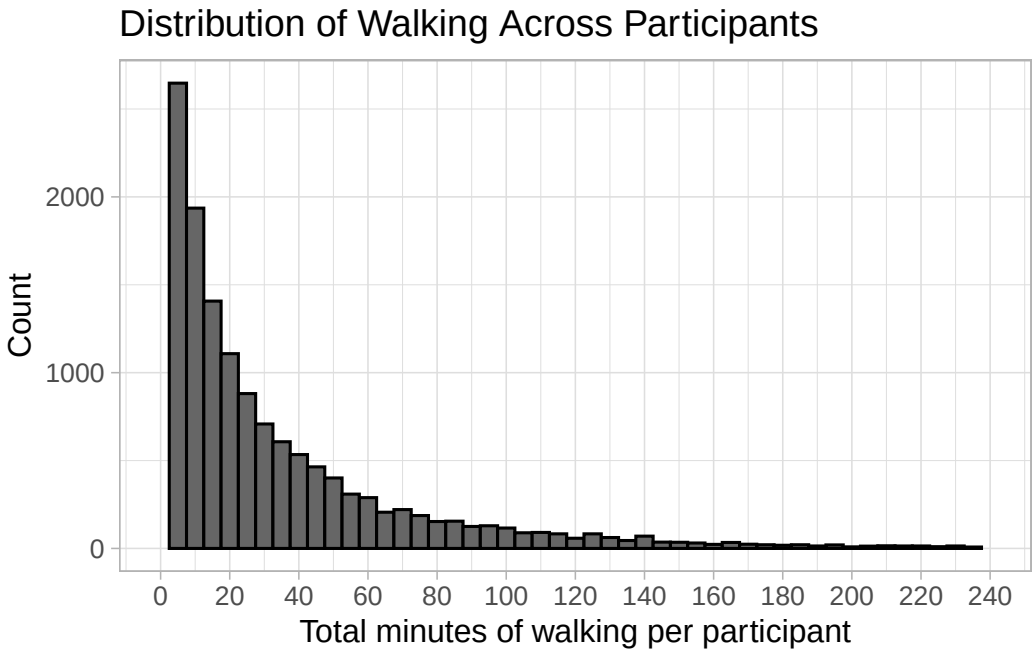
# Nonwear Supplement

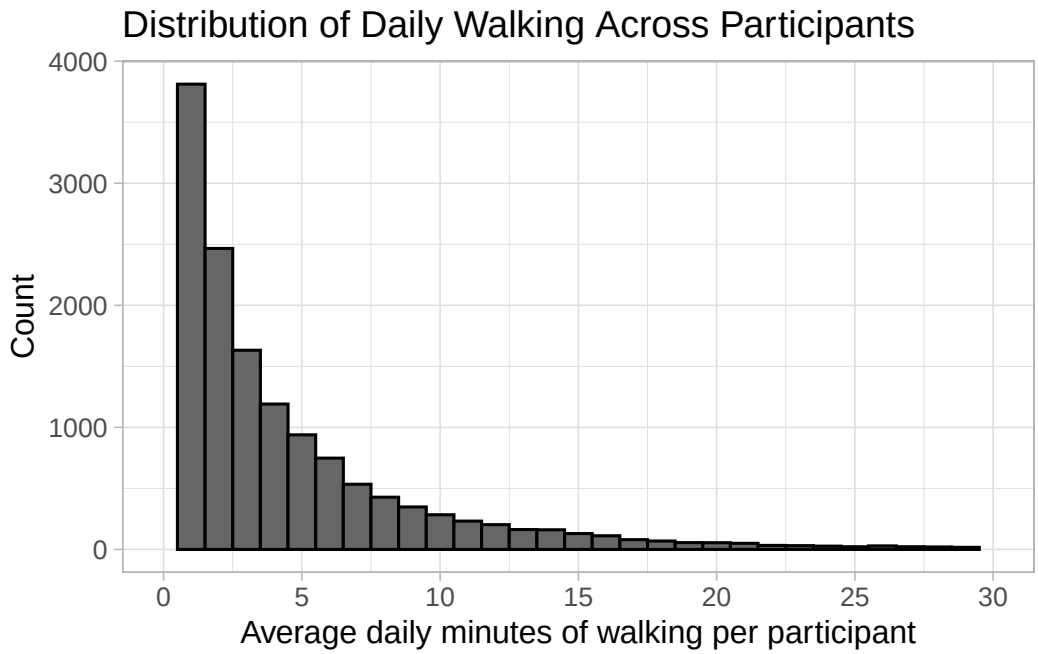
The purpose of this vignette is to demonstrate the amount of ADEPT-labeled walking and nonwear in the data, and investigate whether amount of nonwear may be associated with identification accuracy in our models.

### Amount of walking



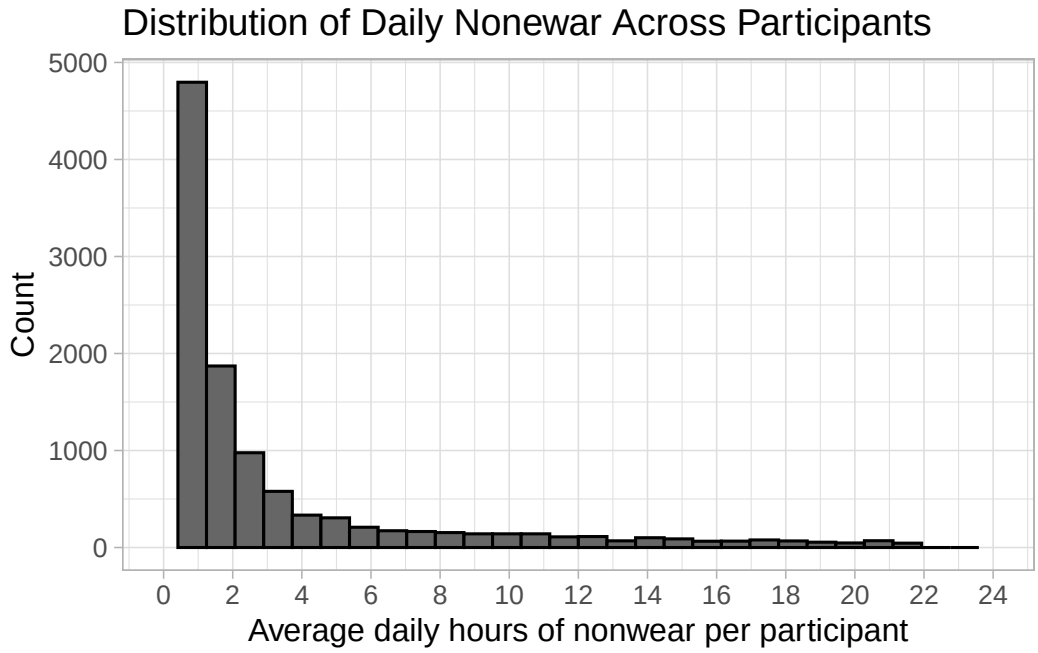

Here walking is defined as a bout of at least 10 seconds of steps, with no more than two seconds between steps.

The median [IQR] number of minutes (across all days) per person is 16.83 [5.87, 41.55].



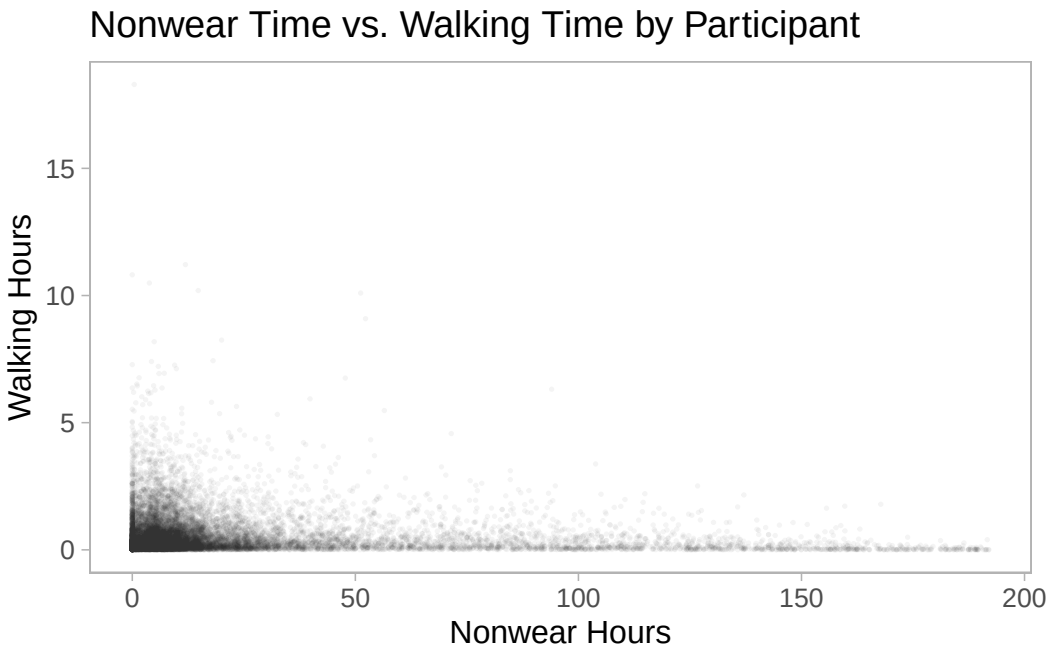

The median [IQR] number of average minutes per person per day is 2.45 [1.05, 5.57]

**Amount of nonwear**



The median [IQR] average number of daily hours of nonwear per person is 0.99 [0.41, 2.59].

### Nonwear and walking



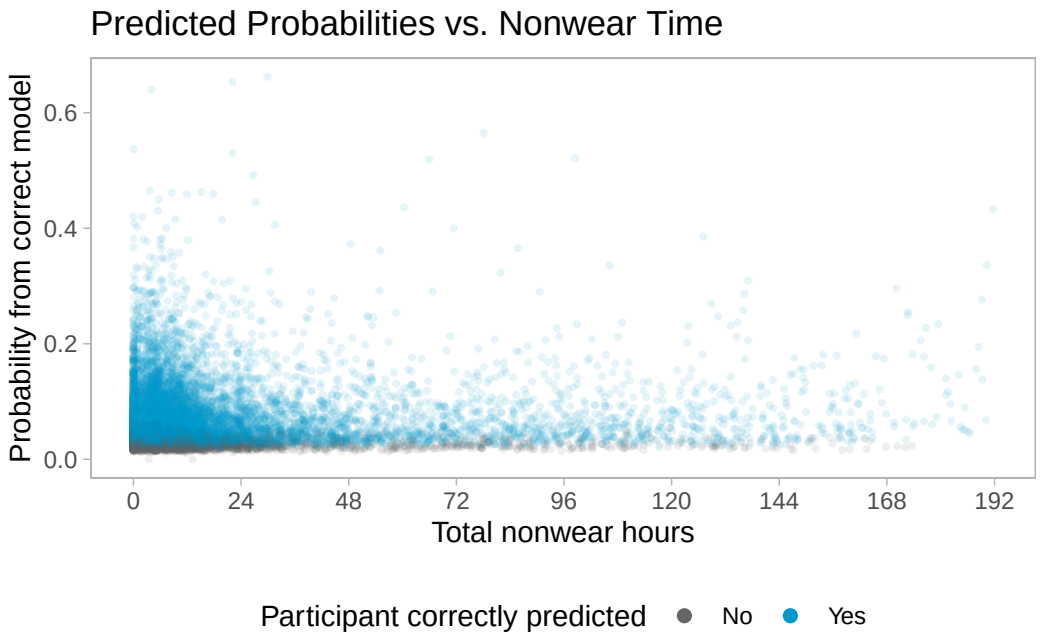

```
model = lm(nonwear_mins ~ walk_mins, data = nw_walk)
broom::tidy(model)
```

```
# A tibble: 2 x 5
  term        estimate std.error statistic  p.value
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)  1550.      22.0        70.6 0
2 walk_mins      -5.22     0.367     -14.2 1.47e-45
```

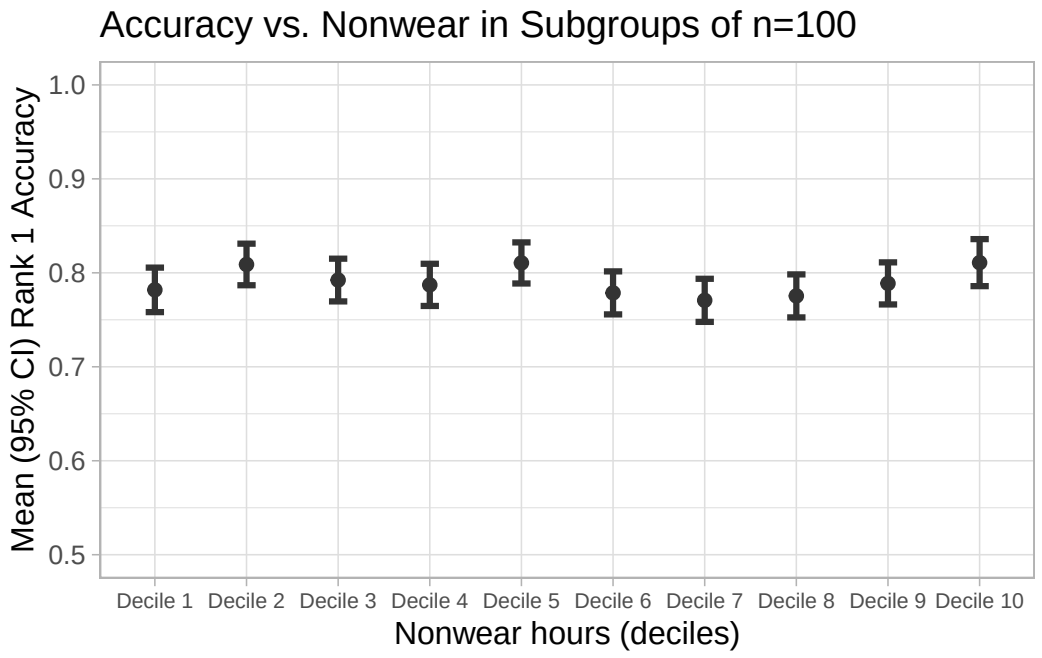
In general, individuals with more nonwear have fewer walking minutes (which we might expect).

### Nonwear and identification accuracy

We can look at identification accuracy in models with $n = 100$ subsets vs. nonwear hours.




The correctly predicted participants seem to occur at similar frequencies across all values of nonwear. We can investigate further by binning nonwear into deciles and looking at accuracy within each decile.




There doesn't appear to be a relationship between nonwear and accuracy. Finally, we can fit a

model where the outcome is whether the individual was correctly predicted and the predictor is nonwear minutes:

```
# A tibble: 2 x 5
  term        estimate std.error statistic p.value
  <chr>          <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)     3.68 0.0266        48.9    0
2 nonwear_min     1.00 0.0000122      1.48   0.138
```

Nonwear time is not significantly associated with correct prediction for a participant.